

Structure-Semantic Synergized Deep Contrastive Graph Clustering

Shifei Ding

School of Computer Science and Technology / School of
Artificial Intelligence, China University of Mining and
Technology
Mine Digitization Engineering Research Center of the
Ministry of Education
Jiangsu Provincial Industrial Technology Engineering
Center for Intelligent Sensing and Emergency IoT in
Underground Space
Xuzhou, Jiangsu, China
dingsf@cumt.edu.cn

Xiao Xu*

School of Computer Science and Technology / School of
Artificial Intelligence, China University of Mining and
Technology
Mine Digitization Engineering Research Center of the
Ministry of Education
Jiangsu Provincial Industrial Technology Engineering
Center for Intelligent Sensing and Emergency IoT in
Underground Space
Xuzhou, Jiangsu, China
xu_xiao@cumt.edu.cn

Zhe Li

School of Computer Science and Technology / School of
Artificial Intelligence, China University of Mining and
Technology
Xuzhou, Jiangsu, China
zhelicumt@cumt.edu.cn

Chao Li

School of Computer Science and Technology / School of
Artificial Intelligence, China University of Mining and
Technology
Xuzhou, Jiangsu, China
chaoli@cumt.edu.cn

Abstract

Current contrastive graph clustering approaches suffer from insufficient integration of structural and semantic information, coupled with the absence of reliable sample selection strategies. To address these dual limitations, we introduce Structure-Semantic Synergized Deep Contrastive Graph Clustering (S³-DCGC), a novel framework that jointly models topological structure and semantic features through two synergistic mechanisms. First, a structure-aware curriculum negative sampling strategy progressively identifies hard negative samples using dynamic-range masking, enhancing discriminative structural learning. Second, a semantic confidence-guided contrastive mechanism quantifies node reliability via composite confidence scores—integrating cluster affinity and cross-view consistency—to select high-confidence positive/negative pairs. Dynamically coordinated by a soft-alignment strategy that shifts optimization focus from structural to semantic dominance during training, these components achieve balanced synergy. Comprehensive experiments conducted on five benchmark datasets demonstrate S³-DCGC's superiority, achieving significant performance gains.

*Corresponding author



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Ablation studies and visual analyses further corroborate the critical importance of structure-semantic synergy in achieving robust clustering performance.

CCS Concepts

• Computing methodologies → Neural networks.

Keywords

Graph Clustering, Contrastive Learning, Graph Neural Networks

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1 Introduction

Graph clustering, as a fundamental task in graph data analysis, aims to partition the nodes of a graph into several semantically coherent clusters based on their structural relationships and semantic features. With the advancement of deep learning, deep graph clustering methods have widely integrated Graph Neural Networks (GNNs) and autoencoder architectures[3, 19, 32]. This integration enables the joint modeling of complex nonlinear structures and attribute information within graphs, leading to significant performance improvements in unsupervised graph representation learning. Contrastive learning, an emerging paradigm for unsupervised

representation learning, has also gained considerable attention in graph clustering tasks in recent years. It effectively enhances the discriminative power of the feature space by maximizing the consistency between positive sample pairs across different views.

However, existing contrastive learning-based graph clustering methods often suffer from two limitations: they overlook intrinsic node information in semantic space during embedding acquisition, failing to synergistically leverage the complementary concepts of structure and semantics, and their contrastive learning lacks guidance in constructing positive/negative samples. To address these issues, we propose a Structure-Semantic Synergized Deep Contrastive Graph Clustering (S^3 -DCGC), which jointly models structural awareness and semantic adaptiveness across two dimensions to guide a more stable and robust contrastive learning process. At the structural level, a structure-aware negative sample mining mechanism amplifies topological disparities; at the semantic level, a cluster center distance-based semantic confidence-guided mechanism filters and weights semantically consistent positive samples to enhance their quality and contrastive effectiveness. Furthermore, a soft alignment mechanism is introduced to achieve phased integration of structural awareness and semantic alignment during training, enabling the acquisition of more stable and higher-quality clustering representations under unsupervised conditions. Our main contributions are as follows:

- A dual-encoder network employing MLP encoders on spectrally-smoothed features to construct semantically diverse node embeddings for contrastive learning.
- A novel structure-semantic synergized deep contrastive graph clustering framework for joint modeling of structural relationships and semantic features.
- Extensive experiments on multiple real-world graph datasets demonstrating the method's superiority in clustering performance and robustness.

2 RELATED WORK

Clustering is a challenging task that learns semantic representations and graph structure in order to assign them into different clusters [9, 16, 23, 29]. In recent years, deep learning methods have gained increasing attention in this field [12, 28], among which deep graph clustering has become a particularly active research area. GAE [13] first employed an inner product decoder for structure restoration. DAEGC [26] introduced attention mechanisms into the encoder. SDCN [2] fused structure and attribute features via dual encoders. AGCN [22] and CaEGCN [11] further explored dynamic and cross-attention fusion strategies. MBN [30] enhanced the decoder and treated GCN as part of the encoder. GC-SEE [6] combined embeddings from diverse encoders for richer representations. Fast-DGC [5] addressed representation collapse with a dynamic graph updating strategy. With the rapid development of deep graph clustering, contrastive learning has become a dominant paradigm due to its ability to capture intrinsic structure and semantics without supervision. To further enhance clustering performance, recent efforts have explored more fine-grained strategies from both structural and semantic perspectives.

2.1 Structure-aware Contrastive Graph Clustering

Existing contrastive graph clustering methods often exploit graph topology to construct meaningful contrastive pairs. For example, MVGRL [10] generates structural views by applying diffusion operators to the adjacency matrix and aligns node-level representations across original and diffused graphs. DCRN [15] uses edge perturbation and structural regularization to enhance structural representation. SCGC [17] builds structure-aware positive pairs by identifying neighbors within the same cluster, leveraging structural consistency. However, these approaches either ignore the reliability of individual nodes or fail to dynamically adjust to the learning stage, making them susceptible to noisy topology or over-smoothing. Our method adopts a curriculum negative sampling strategy that incorporates structural sensitivity into the selection process, enabling adaptive curriculum learning in the structural view.

2.2 Semantic-guided Sample Selection

Semantic information plays a crucial role in contrastive learning, especially when structural cues are ambiguous. Several works, such as CCGC [31] and HSAN [18], utilize pseudo-labels or high-confidence clustering results to select semantically consistent positive pairs. MGCCN [14] aligns cross-view embeddings and uses confident pairs to guide learning. However, most of these strategies rely heavily on fixed thresholds or global pseudo-labels, lacking fine-grained control over sample quality. In contrast, our method computes semantic confidence at node level by combining intra-cluster compactness and cross-view consistency, enabling a more precise and dynamic identification of reliable samples.

2.3 Structure-Semantic Collaboration in Contrastive Clustering

Despite recent progress in both structure- and semantic-guided contrastive learning, few methods consider their joint optimization. Models such as SDCN [2] and EFR-DGC [8] usually apply naive fusion or treat structure and semantics independently. To bridge this gap, we propose a structure-semantic collaborative mechanism that performs joint sample selection and dynamically balances structural and semantic contributions. Specifically, we integrate a structure-aware curriculum negative sampling with a semantic confidence-based positive pair selector. A soft-balancing function is further introduced to coordinate their influence throughout training. This joint design leads to more robust representation learning and superior clustering performance, as verified by extensive experiments.

3 The Proposed Method

3.1 Notations Definition

Given an undirected graph $\mathcal{G}$ with $\mathcal{E}$ edges and N nodes belonging to k categories. $\mathbf{X}$ is the attribute matrix. $\mathbf{A}$ is the adjacency matrix of the graph. $\mathbf{I}$ is the identity matrix. $\hat{\mathbf{A}}$ is the adjacency matrix with self-loops, defined as $\hat{\mathbf{A}} = \mathbf{A} + \mathbf{I}$. $\mathbf{D}$ is the degree matrix of the graph, which is a diagonal matrix with $d_i = \sum_{j=1}^n \mathbf{A}_{ij}$. $\mathbf{L}$ is the Laplacian matrix of the graph, defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$, and the symmetrically

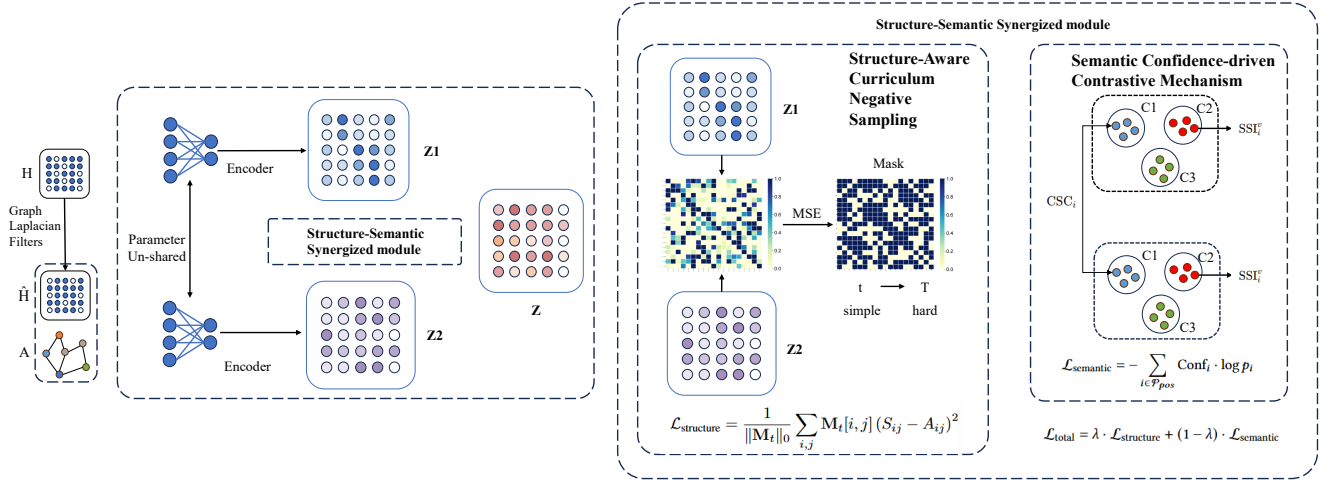


Figure 1: Illustration of our Structure-Semantic Synergized Deep Contrastive Graph Clustering (S³-DCGC). It has two main components: a Dual-Encoder Network that denoises input node features and fuses embeddings, and a Structure-Semantic Synergized module.

normalized Laplacian matrix is $\tilde{L} = \hat{D}^{-\frac{1}{2}} \hat{L} \hat{D}^{-\frac{1}{2}}$. Z_v is the attribute embeddings in v -th view. z_i^v is the embedding of node i in v -th view.

3.2 Overall Framework

We propose a Structure-Semantic Synergized Deep Contrastive Graph Clustering (S³-DCGC). The framework of our module is shown in Figure 1. To more effectively address the complex and diverse distributions of graph data in real-world scenarios, we propose a Structure-Semantic Synergized Deep Contrastive Graph Clustering (S³-DCGC). This framework aims to fully exploit both the semantic attributes of nodes and their topological structural information within graphs, thereby achieving more discriminative and robust graph representation learning.

3.3 Dual-Encoder Network

To enhance the diversity of node representations and mitigate view collapse, following SCGC, we design a dual-encoder network where each encoder independently transforms the input features without parameter sharing. This asymmetric structure enables the model to capture distinct latent patterns from the same input, effectively simulating multi-view observations. Before feature extraction, we apply spectral-based smoothing to incorporate local structural context. Given an input feature matrix X and the symmetrically normalized Laplacian matrix $\tilde{L}$, the smoothed representation is defined as:

$$X^{\text{smooth}} = (I - \tilde{L})^m X, \quad (1)$$

where m controls the extent of propagation across the graph. This preprocessing step encourages the encoder to perceive neighborhood-aware semantics.

We then employ two separate MLP-based encoders, denoted as $f_1(\cdot)$ and $f_2(\cdot)$, to produce two distinct embeddings:

$$Z_1 = f_1(X^{\text{smooth}}), \quad Z_2 = f_2(X^{\text{smooth}}), \quad (2)$$

where Z_1 and Z_2 represent the embeddings from two contrasting views. While the architecture of f_1 and f_2 is kept identical, their weights are independently optimized, ensuring representational discrepancy.

3.4 Structure-aware Curriculum Negative Sampling

Maintaining consistency in structural information between nodes is crucial for learning high-quality graph embeddings in deep graph clustering tasks. Therefore, we propose a structure-aware curriculum negative sampling to guide the model in progressively learning discriminative information within the structure, particularly the ability to distinguish hard negative samples.

We first obtain node embeddings $Z_1 \in R^{N \times d}$ and $Z_2 \in R^{N \times d}$ from our dual-encoder network, where N denotes the number of nodes and d the embedding dimension. To preserve graph structure consistency across views, we compute the cross-view node embedding similarity matrix:

$$S = Z_1 \cdot Z_2^T \in R^{N \times N} \quad (3)$$

Using the adjacency matrix $A \in \{0, 1\}^{N \times N}$ as supervision signal ($A_{ij} = 1$ indicates an edge between nodes i and j), we minimize the Mean Squared Error (MSE) between S and A to preserve structural relationships:

$$\mathcal{L}_{\text{structure}} = \frac{1}{N^2} \sum_{i,j=1}^N (S_{ij} - A_{ij})^2 \quad (4)$$

This loss treats connected node pairs ($A_{ij} = 1$) as positive samples and non-connected pairs ($A_{ij} = 0$) as negative samples. However, edge sparsity in real-world graphs causes severe positive-negative sample imbalance, where excessive irrelevant negative samples may obscure effective structural learning and destabilize training.

To address this, we introduce a curriculum learning mechanism that progressively filters negative samples, constructing a training set of incrementally increasing difficulty to guide structural learning from easy to hard samples. The core innovation is an iteration-dependent mask matrix that gradually expands the "focus interval" during training, selectively including only more "reliable" or "easier" negative samples.

Specifically, at training epoch t , we define a sampling interval $[e(t - c), e(t)]$ following DCLN[21]. Here $e(t)$ denotes a dynamic upper-bound function that expands with epochs, c is a constant controlling initial difficulty (e.g., $c = 10$), and $e(t)$ is formulated as:

$$e(t) = \left(\frac{t}{T}\right)^\gamma (N - 1) \quad (5)$$

where T is the total training epochs and γ a hyperparameter controlling growth rate. We uniformly sample s node-pair indices $\mathcal{I}_{\text{neg}}$ from this interval to construct an epoch-specific mask matrix $\mathbf{M}_t$:

$$\mathbf{M}_t[i, j] = \begin{cases} 1, & \text{if } (i, j) \in \mathcal{I}_{\text{neg}} \text{ and } \mathbf{A}_{ij} = 0 \\ 1, & \text{if } \mathbf{A}_{ij} = 1 \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

Multiplying this mask with the similarity matrix preserves selected negatives and all positives:

$$\mathbf{S}_{\text{masked}} = \mathbf{S} \odot \mathbf{M}_t \quad (7)$$

The final loss is updated to:

$$\mathcal{L}_{\text{structure}} = \frac{1}{\|\mathbf{M}_t\|_0} \sum_{i,j} \mathbf{M}_t[i, j] (\mathbf{S}_{ij} - \mathbf{A}_{ij})^2 \quad (8)$$

This mechanism initially focuses on positive samples and structurally easy negatives, then progressively incorporates harder negatives in later stages, effectively mitigating structural noise interference while enhancing embedding discriminability and training stability.

3.5 Semantic Confidence-guided Contrastive Mechanism

To enhance semantic consistency and reliability in cross-view contrastive learning, we propose a Semantic Confidence-guided Contrastive Mechanism (SCCM). This mechanism leverages the original semantic embeddings without introducing additional semantic enhancement modules by applying static K-means clustering and semantic consistency evaluation to select reliable positive and negative sample pairs from multiple perspectives. This strategy helps improve the discriminative power and robustness of graph clustering.

Given node embeddings in two views, denoted as $\mathbf{z}_i^1$ and $\mathbf{z}_i^2$, we perform view-specific K-means clustering with a predefined number of clusters K , yielding pseudo-label assignments:

$$\mathcal{C}^v = \{c_i^v\}_{i=1}^N, \quad v \in \{1, 2\} \quad (9)$$

Here, $c_i^v \in \{1, \dots, K\}$ indicates the cluster assignment of node i in view v , and $\boldsymbol{\mu}_k^v$ denotes the centroid of cluster k in that view. These cluster labels serve as the foundation for subsequent confidence computation and sample selection.

To determine the reliability of each node's semantic representation across views, we define a semantic confidence score Conf_i , which jointly captures its semantic clarity and cross-view stability:

Semantic Soft Assignment (SSI) measures a node's degree of affiliation to its cluster centroid in each view. It is computed via a softmax-based distance normalization:

$$\text{SSI}_i^v = \frac{\exp(-\|\mathbf{z}_i^v - \boldsymbol{\mu}_{c_i^v}^v\|_2/\sigma)}{\sum_{k=1}^K \exp(-\|\mathbf{z}_i^v - \boldsymbol{\mu}_k^v\|_2/\sigma)}$$

where σ is a temperature hyperparameter controlling the softness of assignment.

Cross-view Semantic Consistency (CSC) evaluates both the semantic similarity and cluster agreement of a node across views:

$$\text{CSC}_i = \text{sim}(\mathbf{z}_i^1, \mathbf{z}_i^2) \cdot I[c_i^1 = c_i^2] \quad (10)$$

Here, sim denotes cosine similarity, and $I[\cdot]$ is the indicator function, equal to 1 when the cluster assignments match and 0 otherwise. This term penalizes semantic divergence across views.

Then we combine these components into a single semantic confidence score:

$$\text{Conf}_i = \beta \cdot \frac{\text{SSI}_i^1 + \text{SSI}_i^2}{2} + (1 - \beta) \cdot \text{CSC}_i \quad (11)$$

where $\beta \in [0, 1]$ is a hyperparameter controlling the contribution of the average intra-view reliability versus the inter-view consistency.

Based on the confidence scores, we construct the positive sample set $\mathcal{P}_{\text{pos}}$ by selecting cross-view pairs of the same node whose semantic confidence exceeds a certain threshold τ :

$$\mathcal{P}_{\text{pos}} = \{(\mathbf{z}_i^1, \mathbf{z}_i^2) \mid \text{Conf}_i \geq \tau\} \quad (12)$$

The threshold τ can be set dynamically, for example, as the top $p\%$ percentile of all confidence scores, to ensure only semantically reliable samples are included.

To construct high-quality negative sample pairs, we focus on the semantic separation between cluster centroids across views. For every pair of distinct clusters (i, j) , the semantic distance is defined as:

$$\text{Dist}_{i,j} = \|\boldsymbol{\mu}_i^1 - \boldsymbol{\mu}_j^2\|_2, \quad i \neq j \quad (13)$$

We select the most distant cluster pairs, and from their corresponding node sets, we extract nodes with confidence scores above a threshold τ to form the negative sample set:

$$\mathcal{P}_{\text{neg}} = \{(\mathbf{z}_p^1, \mathbf{z}_q^2) \mid c_p^1 = i, c_q^2 = j, \text{Conf}_p, \text{Conf}_q \geq \tau\} \quad (14)$$

This approach ensures that the negative samples are semantically dissimilar and internally consistent.

Finally, we define a semantic confidence-weighted contrastive loss that emphasizes the contribution of reliable positive pairs while effectively separating them from semantically confident negatives.

For each selected positive pair $(\mathbf{z}_i^1, \mathbf{z}_i^2) \in \mathcal{P}_{\text{pos}}$, we compute its positive similarity score as:

$$s_i^+ = \exp\left(\frac{\text{sim}(\mathbf{z}_i^1, \mathbf{z}_i^2)}{\tau_t}\right) \quad (15)$$

Similarly, we define the negative similarity accumulation based on its associated negative set $\mathcal{P}_i^-$:

$$s_i^- = \sum_{(z_i^1, z_j^2) \in \mathcal{P}_i^-} \exp\left(\frac{\text{sim}(z_i^1, z_j^2)}{\tau_t}\right) \quad (16)$$

Based on these, the semantic contrastive probability for the positive pair is computed as:

$$p_i = \frac{s_i^+}{s_i^+ + s_i^-} \quad (17)$$

Incorporating the semantic confidence score Conf_i as a soft weighting factor, the final semantic contrastive loss is given by:

$$\mathcal{L}_{\text{semantic}} = - \sum_{i \in \mathcal{P}_{\text{pos}}} \text{Conf}_i \cdot \log p_i \quad (18)$$

Here, τ_t is a temperature scaling factor that controls the sharpness of the similarity distribution. This formulation reinforces high-confidence semantic alignment and effectively suppresses noisy or ambiguous samples, thereby enhancing both the stability and discriminability of the learned embeddings.

3.6 Synergistic Optimization

Building upon the structure-aware curriculum negative sampling and semantic confidence-guided contrastive mechanisms, we integrate these complementary perspectives through a unified learning framework. The structure-aware mechanism focuses on local structural similarity by leveraging adjacency relationships and sampling strategies to construct negative samples, while the semantic confidence mechanism exploits global semantic distributions and clustering consistency to select positive and negative pairs, thereby enhancing semantic discriminability in cross-view contrastive learning.

To achieve synergistic optimization of these dual perspectives, we formulate the composite loss function:

$$\mathcal{L}_{\text{total}} = \lambda \cdot \mathcal{L}_{\text{structure}} + (1 - \lambda) \cdot \mathcal{L}_{\text{semantic}} \quad (19)$$

where $\mathcal{L}_{\text{structure}}$ denotes the contrastive loss from the structure-aware curriculum sampling module, $\mathcal{L}_{\text{semantic}}$ represents the confidence-weighted contrastive loss from the semantic mechanism, and $\lambda \in [0, 1]$ serves as the balancing factor between structural and semantic components.

We further introduce a soft alignment strategy to dynamically modulate λ across training epochs t :

$$\lambda(t) = 1 - \frac{1}{1 + \exp(-\alpha(t - t_0))} \quad (20)$$

with hyperparameter α controlling the transition speed and t_0 determining the critical epoch where semantic learning begins to dominate. This strategy implements a local-to-global, easy-to-hard curriculum: structure-centric learning ($\lambda \approx 1$) predominates in early stages, gradually transitioning to semantics-centric optimization ($\lambda \approx 0$) in later phases, thereby aligning with fundamental cognitive learning principles while ensuring stable convergence.

4 Experimental

4.1 Experimental Environment

- **CPU:** R9-7940HX
- **GPU:** NVIDIA GeForce RTX 4060 GPU, 16GB RAM
- **Deep Learning Platform:** PyTorch

Algorithm 1 Structure-Semantic Synergized Deep Contrastive Graph Clustering (S^3 -DCGC)

Input: Graph $\mathcal{G} = \{\mathbf{X}, \mathbf{A}\}$; cluster number C ; total iteration I ; warm-up epoch T_w ; GNN layer number m ; Gaussian noise std η .

Output: Clustering result $\mathbf{R}$

```

Encode raw node features  $\mathbf{X}$  using an autoencoder to obtain semantic embedding  $\mathbf{Z}_1$ .
Apply GNN encoder with  $m$ -layer Laplacian smoothing to get structural embedding  $\mathbf{Z}_2$ .
Fuse two views to get representation  $\mathbf{Z} = 1/2(\mathbf{Z}_1 + \mathbf{Z}_2)$ .
for  $i = 1$  to  $I$  do
    Update cluster centers via K-means over  $\mathbf{Z}$ .
    Generate results based on current cluster assignment.
    Early training phase performs structure-aware curriculum negative sampling through Eq.8.
    Later training phase selects positive pairs guided by semantic confidence through Eq.18.
    Compute soft-balancing weight  $\lambda$  based on current epoch.
    Compute total loss Eq.19.
end for
Obtain final clustering result  $\mathbf{R}$  via K-means on  $\mathbf{Z}$ .
return  $\mathbf{R}$ 

```

Our model is trained by optimizing Eq.19 for 400 epochs. The detailed learning procedure is shown in Algorithm 1. We replicate the results using the original settings and source code. The optimizer's learning rates are set as follows: $1e-4$ for CORA, BAT, AMAP, CITESEER and UAT. The standard deviation η of the random Gaussian noise is set to 0.01. The threshold τ is set as follows: 0.8 for CORA, UAT and BAT, 0.9 for CITESEER, 0.7 for AMAP. The parameter of the loss function t_0 is set as follows: 100 for CORA, 200 for CITESEER and AMAP, 300 for BAT, 50 for UAT.

4.2 Models for Comparison

- **MGAE[27]:** MGAE proposes a marginalized graph autoencoder that corrupts node content to interact with network features, learning effective representations for spectral clustering.
- **DAEGC[26]:** DAEGC exploits an attentional neighbor-wise strategy to encode node representations, supervised by a specific clustering loss function.
- **ARGA[20]:** ARGA employs a variational graph auto-encoder to capture the data prior distribution and is optimized via an adversarial training process involving a generator and a discriminator.
- **SDCN[2]:** SDCN combines GCN with the DEC framework to integrate topological structural information into the deep clustering process.
- **DFCN[24]:** DFCN fuses embeddings from an autoencoder and a graph autoencoder via self-correlation to capture global-local information for self-supervised clustering.

Table 1: Clustering results on five datasets. (The best result is highlighted with bold.)

Dataset	Metric	MGAE	DAEGC	ARGA	SDCN	DFCN	AGE	MVGRL	AGC-DRR	SCGC	CCGC	OURS
CORA	ACC	43.38±2.11	70.43±0.36	71.04±0.25	35.60±2.83	36.33±0.49	73.50±1.83	70.47±3.70	40.62±0.55	73.88±0.88	73.88±1.20	76.10±0.77
	NMI	28.78±2.97	52.89±0.69	51.06±0.52	14.28±1.91	19.36±0.87	57.58±1.42	55.57±1.54	18.74±0.73	56.10±0.72	56.45±1.04	57.07±1.03
	ARI	16.43±1.65	49.63±0.43	47.71±0.33	07.78±3.24	04.67±2.10	50.10±2.14	48.70±3.94	14.80±1.64	51.79±1.59	52.51±1.89	54.61±0.93
	F1	33.48±3.05	68.27±0.57	69.27±0.39	24.37±1.04	26.16±0.50	69.28±1.59	67.15±1.86	31.23±0.57	70.81±1.96	70.98±2.79	74.29±0.94
CITE	ACC	61.35±0.80	64.54±1.39	61.07±0.49	65.96±0.31	69.50±0.20	69.73±0.24	62.83±1.59	68.32±1.83	71.02±0.77	69.84±0.94	71.55±0.35
	NMI	34.63±0.65	36.41±0.86	34.40±0.71	38.71±0.32	43.90±0.20	44.93±0.53	40.69±0.93	43.28±1.41	45.25±0.45	44.33±0.79	45.45±0.37
	ARI	33.55±1.18	37.78±1.24	34.32±0.70	40.17±0.43	45.50±0.30	45.31±0.41	34.18±1.73	45.34±2.33	46.29±1.13	45.68±1.80	46.62±0.81
	F1	57.36±0.82	62.20±1.32	58.23±0.31	63.62±0.24	64.30±0.20	64.45±0.27	59.54±2.17	64.82±1.60	64.80±1.01	62.71±2.06	62.36±0.41
AMAP	ACC	71.57±2.48	75.96±0.23	69.28±2.30	53.44±0.81	76.82±0.23	75.98±0.68	41.07±3.12	76.81±1.45	77.48±0.37	77.25±0.41	78.36±0.59
	NMI	62.13±2.79	65.25±0.45	58.36±2.76	44.85±0.83	66.23±1.21	65.38±0.61	30.28±3.94	66.54±1.24	67.67±0.88	67.44±0.48	67.35±0.51
	ARI	48.82±4.57	58.12±0.24	44.18±4.41	31.21±1.23	58.28±0.74	55.89±1.34	18.77±2.34	60.15±1.56	58.48±0.72	57.99±0.66	59.72±0.67
	F1	68.08±1.76	69.87±0.54	64.80±1.95	50.66±1.49	71.25±0.31	71.74±0.93	32.88±5.50	71.03±0.64	72.22±0.97	72.18±0.57	73.30±2.01
BAT	ACC	53.59±2.04	52.67±0.00	67.36±0.80	53.05±4.63	55.73±0.06	56.64±0.76	37.56±0.32	47.79±0.02	77.97±0.99	75.04±1.78	79.54±0.39
	NMI	30.59±2.06	21.43±0.35	49.09±0.54	25.74±5.71	48.77±0.51	36.08±1.54	29.33±0.70	19.91±0.24	52.91±0.68	50.23±2.43	55.15±1.34
	ARI	24.15±1.70	18.18±0.29	42.02±1.21	21.04±4.97	37.76±0.23	26.59±1.83	13.45±0.03	14.59±0.13	50.64±1.85	46.95±3.09	54.34±0.66
	F1	50.83±3.23	52.23±0.03	67.02±1.15	46.45±5.90	50.90±0.12	55.07±0.80	29.64±0.49	42.33±0.51	78.03±0.96	74.90±1.80	79.56±0.44
UAT	ACC	48.97±1.52	52.29±0.49	49.31±0.15	52.25±1.91	33.61±0.09	52.37±0.42	44.16±1.38	42.64±0.31	56.58±1.62	56.34±1.11	59.53±0.68
	NMI	20.69±0.98	21.33±0.44	25.44±0.31	21.61±1.26	26.49±0.41	23.64±0.66	21.53±0.94	11.15±0.24	28.07±0.71	28.15±1.92	29.95±1.22
	ARI	18.33±1.79	20.50±0.51	16.57±0.31	21.63±1.49	11.87±0.23	20.39±0.70	17.12±1.46	09.50±0.25	24.80±1.85	25.52±2.09	28.06±1.11
	F1	47.95±1.52	50.33±0.64	50.26±0.16	45.59±3.54	25.79±0.29	50.15±0.73	39.44±2.19	35.18±0.32	55.52±0.87	55.24±1.69	59.74±1.34

Table 2: Datasets.

Dataset	Type	Sample	Dimension	Edge	Class
CORA	Graph	2708	1433	5429	7
CITESEER	Graph	3327	3703	4732	6
AMAP	Graph	7650	745	119081	8
BAT	Graph	131	81	1038	4
UAT	Graph	1190	239	13599	4

- **AGE[4]**: AGE utilizes graph Laplacian filters and an adaptive matrix to replace standard graph convolutional networks for efficient feature extraction.
- **MVGRL[10]**: MVGRL incorporates a contrastive learning mechanism into the graph clustering framework, effectively advancing research in contrastive-based clustering.
- **AGC-DRR[7]**: AGC-DRR achieves the reduction of redundant information within both the input features and the latent space representations.
- **SCGC[17]**: SCGC utilizes parameter un-shared Siamese encoders with graph Laplacian filters to generate cross-view embeddings for sample selection.
- **CCGC[31]**: CCGC optimizes contrastive pairing by utilizing a sample selection strategy grounded in high-confidence pseudo labels.

4.3 Benchmark Datasets

We conducted experiments on five widely used datasets[1]: CORA, CITESEER, AMAP, BAT, and UAT. A brief introduction to the datasets is provided in Table 2.

4.4 Performance Comparison

The overall clustering results are shown in Table 1. To demonstrate the superiority of our proposed model, we conduct extensive experiments by comparing it with ten representative baseline methods, including deep autoencoder-based models[2, 20, 24, 27], graph

attention methods[26], and recent contrastive graph clustering model[4, 7, 10, 17, 31]. As shown in the results table, our method achieves a clear advantage on most datasets. When compared with encoder-based methods such as DFCN, our structure-semantic collaborative contrastive learning framework substantially enhances both structural consistency and semantic separability in the learned features.

For models like DAEGC that incorporate attention mechanisms, we further introduce semantic confidence modeling and cross-view alignment strategies to mitigate the noise introduced by attention aggregation.

Compared with SDCN, which fuses structural and attribute information, our method explicitly distinguishes between high- and low-confidence semantic samples, enabling better discrimination near clustering boundaries. Most notably, when compared with recent contrastive clustering models such as CCGC and SCGC, our approach not only achieves better results but also provides a more interpretable and controllable sample construction strategy. These findings collectively validate the effectiveness of our structure-semantic collaborative modeling design.

4.5 Ablation Study

4.5.1 Effectiveness of Structure-Semantic Collaboration. To further demonstrate the effectiveness of the proposed Structure-Semantic Collaboration Module (SSCM), we conduct a comprehensive ablation study by comparing the complete model with its two simplified variants: one without the semantic-guided mechanism (denoted as (w/o)semantic) and the other without the structural-aware contrastive design (denoted as (w/o)structure). The results are reported in Table 3 across five benchmark datasets. We conclude that the model without semantic confidence guidance ((w/o)semantic) suffers a noticeable performance drop, especially on datasets like CORA and UAT. This implies that solely relying on structure without distinguishing semantically reliable samples may introduce noise into contrastive learning. Conversely, removing structural discrimination ((w/o)structure) also leads to a consistent decrease

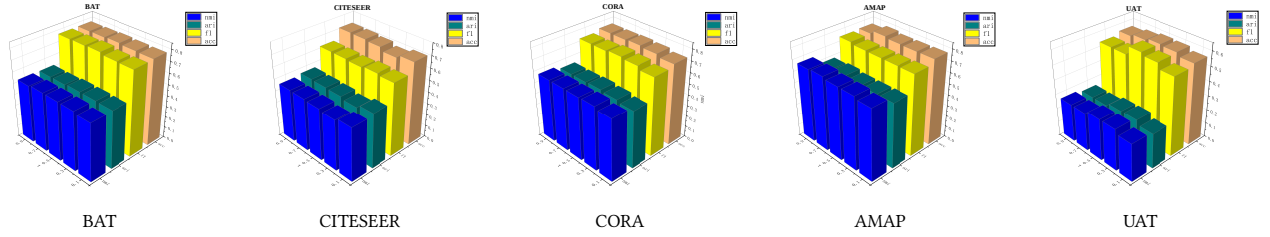
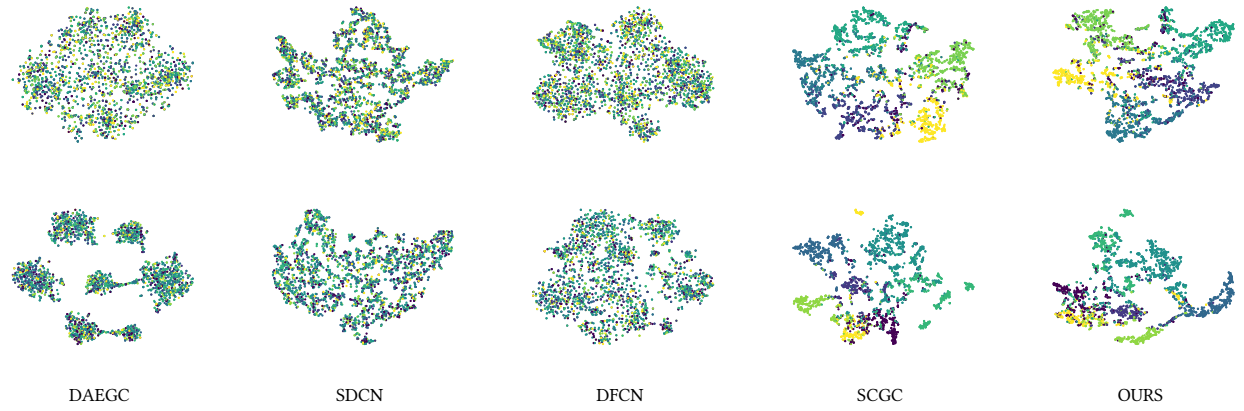
Figure 2: Analysis of the t_0 on five datasets.Figure 3: Analysis of the threshold τ on five datasets

Figure 4: Analysis of 2D visualization on two datasets. The first row and second row correspond to CITESEER and CORA.

across all metrics, indicating that global structural separation, especially among inter-cluster centers—is critical for improving discriminability. The complete SSCM design, which combines structure-aware hard negative mining with semantic high-confidence selection, consistently achieves the best performance, suggesting the

complementary benefits of integrating structure and semantics. Our results suggest that aligning structure and semantics not only

Table 3: The ablation study on five datasets.

dataset	metric	(w/o)CSC	(w/o)SSI	(w/o)structure	(w/o)semantic	ours
cora	acc	75.34 ± 0.41	74.62 ± 0.45	70.63 ± 0.52	72.83 ± 0.83	76.10±0.77
	nmi	56.57 ± 0.43	55.65 ± 0.35	50.56 ± 0.99	52.34 ± 0.81	57.07±1.03
	ari	53.24 ± 0.46	52.18 ± 0.92	50.45 ± 0.72	51.48 ± 1.91	54.61±0.93
cite	f1	73.42 ± 0.96	71.98 ± 0.87	69.52 ± 1.23	71.46 ± 0.67	74.29±0.94
	acc	70.15 ± 0.25	70.34 ± 0.16	68.59 ± 0.97	69.34 ± 0.19	71.55±0.35
	nmi	43.57 ± 0.52	44.42 ± 0.21	41.59 ± 0.60	42.38 ± 0.16	45.45±0.37
bat	ari	44.27 ± 0.37	44.72 ± 0.35	42.35 ± 1.52	43.29 ± 0.24	46.62±0.81
	f1	61.41 ± 0.76	61.56 ± 0.47	58.83 ± 0.43	60.16 ± 0.35	62.36±0.41
	acc	78.47 ± 0.49	76.25 ± 0.47	75.54 ± 0.75	76.52 ± 0.62	79.54±0.39
uat	nmi	54.10 ± 1.46	53.54 ± 0.91	50.38 ± 0.85	51.24 ± 1.21	55.15±1.34
	ari	53.71 ± 1.40	50.56 ± 1.62	49.54 ± 0.93	51.16 ± 1.12	54.34±0.66
	f1	78.84 ± 0.90	77.76 ± 0.96	74.45 ± 0.29	75.16 ± 0.61	79.56±0.44
amap	acc	58.81 ± 0.89	58.26 ± 0.95	53.24 ± 1.32	54.26 ± 1.51	59.53±0.68
	nmi	28.16 ± 0.64	27.17 ± 0.61	23.27 ± 1.78	24.17 ± 1.36	29.95±1.22
	ari	26.67 ± 0.77	27.31 ± 1.50	21.35 ± 2.83	21.61 ± 1.21	28.06±1.11
amap	f1	56.07 ± 0.98	55.67 ± 0.77	52.74 ± 2.65	52.77 ± 1.65	59.74±1.34
	acc	77.87 ± 2.16	77.26 ± 1.79	76.57 ± 3.23	74.42 ± 2.51	78.36±0.59
	nmi	62.25 ± 1.26	62.59 ± 1.77	63.55 ± 1.56	62.25 ± 2.13	67.35±1.51
amap	ari	55.21 ± 1.49	56.21 ± 1.23	56.68 ± 2.09	55.25 ± 2.56	59.72±0.67
	f1	69.66 ± 1.27	70.24 ± 1.37	69.72 ± 1.21	69.62 ± 2.18	73.30±2.01

boosts contrastive supervision but also facilitates more robust clustering representations.

4.5.2 Semantic Confidence. To further investigate the functional contribution of semantic confidence, we decompose this mechanism into two constituent elements: semantic soft-assignment (SSI) operating at node granularity for reliability screening, and cross-view semantic consistency (CSC) enforcing multi-view agreement. Controlled ablation variants—excluding SSI ((w/o)SSI) and excluding CSC ((w/o)CSC)—are systematically evaluated across five benchmark datasets (Table 3). Experimental evidence demonstrates that removal of SSI-based filtering induces marked performance degradation, particularly pronounced on CORA and UAT benchmarks. This deterioration directly originates from the model’s compromised capacity to discard ambiguous samples residing near cluster boundaries, consequently introducing noisy positive pairs and impairing semantic cluster cohesion. Meanwhile, architectures lacking CSC integration exhibit preserved baseline functionality but reveal heightened sensitivity to multi-view perturbations; this vulnerability manifests as inconsistent pseudo-label alignment across views under noise conditions, as empirically observed on UAT and AMAP datasets.

4.6 Hyperparameter Analysis

4.6.1 Analysis of Sensitivity to the threshold τ . To evaluate the robustness of our method with respect to the threshold for selecting positive samples, we conduct a sensitivity analysis on the hyperparameter τ , which controls the minimum semantic confidence required for a sample pair to be regarded as positive. Specifically, we vary τ within a reasonable range and report the clustering performance. As shown in Fig. 3, our model consistently maintains competitive performance across a wide range of τ values, demonstrating strong tolerance to this hyperparameter.

In addition, we observe the following trends: When τ is set too low, more uncertain samples are included as positives, potentially introducing noise and weakening the effectiveness of discriminative learning. Conversely, an overly strict threshold (e.g., $\tau > 0.9$) results in too few positive pairs, thereby reducing the strength of the contrastive supervision. The best performance is achieved when

τ lies in the range of [0.5, 0.7], which strikes a balance between confidence reliability and sample diversity.

4.6.2 Analysis of Hyperparameter t_0 in the Loss Function. To investigate the impact of the transition point in our structure-semantic collaborative framework, we conduct a sensitivity analysis on the hyperparameter t_0 , which determines when the semantic-guided contrastive branch begins to dominate the training process. Our framework balances the structural contrastive loss $\mathcal{L}_{structure}$ and the semantic confidence-guided loss $\mathcal{L}_{semantic}$ via a time-dependent weighting factor $\lambda(t)$, controlled by a sigmoid function centered at t_0 . By varying t_0 from 0 to 400 in increments of 50, we assess the clustering performance across multiple datasets. As shown in Fig. 2, the model exhibits stable and competitive performance over a wide range of t_0 values, reflecting its robustness to the specific timing of fusion. Notably, extremely small t_0 values lead to premature reliance on noisy semantic signals, while overly large values delay semantic refinement, both of which slightly compromise performance. The optimal performance is generally achieved when $t_0 \in [100, 250]$, suggesting that a moderate delay in semantic dominance helps balance local structural guidance and global semantic consistency.

4.7 Visualization Analysis

To demonstrate the superiority of our model, we use the t -SNE [25] to visualize the distribution of embeddings from our model and four baseline models on the CORA and CITESEER. As illustrated in Figure 4, results show that our S³-DCGC reveals the intrinsic clustering structure more effectively than the other baselines.

5 Conclusion

In this paper, we propose a Structure-Semantic Synergized Deep Contrastive Graph Clustering (S³-DCGC) to address the limitations of insufficient information integration and unreliable sample selection in existing methods. Concretely, we first design a dual-encoder network with spectral smoothing to embed node attributes and structural information into semantically diverse latent spaces. Then, a structure-aware curriculum negative sampling strategy is proposed to progressively identify hard negative samples by utilizing a dynamic range masking method, thus enhancing structural discriminative learning. Furthermore, guided by node reliability, we propose a semantic confidence-guided contrastive mechanism to filter high confidence positive and negative pairs by integrating intra-cluster compactness and cross-view consistency. In this manner, the proposed synergistic optimization with soft alignment forces the network to dynamically shift focus from structural perception to semantic comprehension, thus achieving balanced synergy and robust clustering. Experiments on five benchmark datasets demonstrate the effectiveness and superiority of our proposed method. In this work, the confidence thresholds are determined by fixed percentiles, thus one future work is to develop a learnable thresholding strategy that dynamically adjusts based on the clustering uncertainty during training.

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References

- [1] 2023. Hard Sample Aware Network for Contrastive Deep Graph Clustering. 37 (Jun. 2023), 8914–8922. doi:10.1609/aaai.v37i7.26071
- [2] Deyu Bo, Xiao Wang, Chuan Shi, Meiqi Zhu, Emiao Lu, and Peng Cui. 2020. Structural deep clustering network. In *Proceedings of The Web Conference 2020*. 1400–1410.
- [3] Sandro Cavallari, Vincent W. Zheng, Hongyun Cai, Kevin Chen-Chuan Chang, and Erik Cambria. 2017. Learning Community Embedding with Community Detection and Node Embedding on Graphs. In *Proceedings of the 2017 ACM on Conference on Information and Knowledge Management* (Singapore, Singapore) (CIKM '17). Association for Computing Machinery, New York, NY, USA, 377–386. doi:10.1145/3132847.3132925
- [4] Ganqu Cui, Jie Zhou, Cheng Yang, and Zhiyuan Liu. 2020. Adaptive graph encoder for attributed graph embedding. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 976–985.
- [5] Shifei Ding, Benyu Wu, Ling Ding, Xiao Xu, Lili Guo, Hongmei Liao, and Xindong Wu. 2024. Towards Faster Deep Graph Clustering via Efficient Graph Auto-Encoder. *ACM Trans. Knowl. Discov. Data* 18, 8, Article 202 (aug 2024), 23 pages. doi:10.1145/3674983
- [6] Shifei Ding, Benyu Wu, Xiao Xu, Lili Guo, and Ling Ding. 2023. Graph clustering network with structure embedding enhanced. *Pattern Recognition* 144 (2023), 109833. doi:10.1016/j.patcog.2023.109833
- [7] Lei Gong, Sihang Zhou, Wenxuan Tu, and Xinwang Liu. 2022. Attributed Graph Clustering with Dual Redundancy Reduction. In *IJCAI*. 3015–3021.
- [8] Jie Hao and William Zhu. 2022. Deep graph clustering with enhanced feature representations for community detection. *Applied Intelligence* 53, 2 (April 2022), 1336–1349. doi:10.1007/s10489-022-03381-y
- [9] John A Hartigan and Manchek A Wong. 1979. Algorithm AS 136: A k-means clustering algorithm. *Journal of the royal statistical society. series c (applied statistics)* 28, 1 (1979), 100–108.
- [10] Kaveh Hassani and Amir Hosein Khasahmadi. 2020. Contrastive multi-view representation learning on graphs. In *International Conference on Machine Learning*. PMLR, 4116–4126.
- [11] Guangyu Huo, Yong Zhang, Junbin Gao, Boyue Wang, Yongli Hu, and Baocai Yin. 2021. CaEGCN: Cross-attention fusion based enhanced graph convolutional network for clustering. *IEEE Transactions on Knowledge and Data Engineering* 35, 4 (2021), 3471–3483.
- [12] A. K. Jain, M. N. Murty, and P. J. Flynn. 1999. Data clustering: a review. 31, 3 (Sept. 1999), 264–323. doi:10.1145/331499.331504
- [13] Thomas N Kipf and Max Welling. 2016. Variational graph auto-encoders. *arXiv preprint arXiv:1611.07308* (2016).
- [14] Liang Liu, Zhao Kang, Jiajia Ruan, and Xixu He. 2022. Multilayer graph contrastive clustering network. *Information Sciences* 613 (2022), 256–267. doi:10.1016/j.ins.2022.09.042
- [15] Yue Liu, Wenxuan Tu, Sihang Zhou, Xinwang Liu, Linxuan Song, Xihong Yang, and En Zhu. 2022. Deep graph clustering via dual correlation reduction. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 36. 7603–7611.
- [16] Yue Liu, Jun Xia, Sihang Zhou, Xihong Yang, Ke Liang, Chenchen Fan, Yan Zhuang, Stan Z Li, Xinwang Liu, and Kunlun He. 2022. A Survey of Deep Graph Clustering: Taxonomy, Challenge, Application, and Open Resource. *arXiv preprint arXiv:2211.12875* (2022).
- [17] Yue Liu, Xihong Yang, Sihang Zhou, Xinwang Liu, Siwei Wang, Ke Liang, Wenxuan Tu, and Liang Li. 2023. Simple Contrastive Graph Clustering. *IEEE Transactions on Neural Networks and Learning Systems* (2023), 1–12. doi:10.1109/TNNLS.2023.3271871
- [18] Yue Liu, Xihong Yang, Sihang Zhou, Xinwang Liu, Zhen Wang, Ke Liang, Wenxuan Tu, Liang Li, Jingcan Duan, and Cancan Chen. 2023. Hard sample aware network for contrastive deep graph clustering. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 37. 8914–8922.
- [19] Nairouz Mrabah, Mohamed Bouguessa, Mohamed Fawzi Touati, and Riadh Ksantini. 2023. Rethinking Graph Auto-Encoder Models for Attributed Graph Clustering. *IEEE Transactions on Knowledge and Data Engineering* 35, 9 (2023), 9037–9053. doi:10.1109/TKDE.2022.3220948
- [20] Shirui Pan, Ruiqi Hu, Sai-fu Fung, Guodong Long, Jing Jiang, and Chengqi Zhang. 2019. Learning graph embedding with adversarial training methods. *IEEE transactions on cybernetics* 50, 6 (2019), 2475–2487.
- [21] Xin Peng, Jieren Cheng, Xiangyan Tang, Jingxin Liu, and Jiahua Wu. 2024. Dual Contrastive Learning Network for Graph Clustering. *IEEE Transactions on Neural Networks and Learning Systems* 35, 8 (2024), 10846–10856. doi:10.1109/TNNLS.2023.3244397
- [22] Zhihao Peng, Hui Liu, Yuheng Jia, and Junhui Hou. 2021. Attention-driven Graph Clustering Network. In *Proceedings of the 29th ACM International Conference on Multimedia*. 935–943.
- [23] Yazhou Ren, Jingyu Pu, Zhimeng Yang, Jie Xu, Guofeng Li, Xiaorong Pu, Philip S. Yu, and Lifang He. 2024. Deep Clustering: A Comprehensive Survey. *IEEE Transactions on Neural Networks and Learning Systems* (2024), 1–21. doi:10.1109/TNNLS.2024.3403155
- [24] Wenxuan Tu, Sihang Zhou, Xinwang Liu, Xifeng Guo, Zhiping Cai, Jieren Cheng, et al. 2020. Deep Fusion Clustering Network. *arXiv preprint arXiv:2012.09600* (2020).
- [25] Laurens van der Maaten and Geoffrey Hinton. 2008. Visualizing Data using t-SNE. *Journal of Machine Learning Research* 9, 86 (2008), 2579–2605. <http://jmlr.org/papers/v9/vandermaten08a.html>
- [26] Chun Wang, Shirui Pan, Ruiqi Hu, Guodong Long, Jing Jiang, and Chengqi Zhang. 2019. Attributed graph clustering: a deep attentional embedding approach. In *Proceedings of the 28th International Joint Conference on Artificial Intelligence* (Macao, China) (IJCAI'19). AAAI Press, 3670–3676.
- [27] Chun Wang, Shirui Pan, Guodong Long, Xingquan Zhu, and Jing Jiang. 2017. Mgae: Marginalized graph autoencoder for graph clustering. In *Proceedings of the 2017 ACM on Conference on Information and Knowledge Management*. 889–898.
- [28] Junyuan Xie, Ross Girshick, and Ali Farhadi. 2016. Unsupervised Deep Embedding for Clustering Analysis. In *Proceedings of The 33rd International Conference on Machine Learning* (Proceedings of Machine Learning Research, Vol. 48). Maria Florina Balcan and Kilian Q. Weinberger (Eds.). PMLR, New York, New York, USA, 478–487. <https://proceedings.mlr.press/v48/xieb16.html>
- [29] Rui Xu and Donald Wunsch. 2005. Survey of clustering algorithms. *IEEE Transactions on neural networks* 16, 3 (2005), 645–678.
- [30] Xiaoqiang Yan, Xiangyu Yu, Shizhe Hu, and Yangdong Ye. 2023. Mutual Boost Network for attributed graph clustering. *Expert Systems with Applications* 229 (2023), 120479.
- [31] Xihong Yang, Yue Liu, Sihang Zhou, Siwei Wang, Wenxuan Tu, Qun Zheng, Xinwang Liu, Liming Fang, and En Zhu. 2023. Cluster-guided contrastive graph clustering network. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 37. 10834–10842.
- [32] Xiaotong Zhang, Han Liu, Qimai Li, and Xiao-Ming Wu. 2019. Attributed graph clustering via adaptive graph convolution. In *Proceedings of the 28th International Joint Conference on Artificial Intelligence*. 4327–4333.